Prince Agyei Tuffour

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SUMMARY

Results- driven Software Engineer with a Master of Science in Mathematics, specializing in advanced optimization and facial recognition, now seeking a Software Engineer, Backend (All Teams) role. Proven expertise in architecting scalable platforms, leading cross-functional teams to enhance product quality, and driving significant revenue growth through robust backend development and service-oriented architecture. Adept at leveraging DevOps principles and infrastructure as code to optimize system performance and ensure high availability, with a strong background in machine learning deployment, RESTful API design, and comprehensive unit testing.

EDUCATION

Oregon State University | Corvallis OR
Master of Science | Mathematics

Kwame Nkrumah University of Science and Technology | Ghana
Bachelor of Science | Mathematics

EXPERIENCE

Software QA Engineer | dynaConnections Corporations March 2025 - Present

- Elevated product reliability from 73% to 95% by implementing comprehensive test automation frameworks covering 85% of critical user journeys, reducing post-deployment bugs by 60% to under 60 monthly incidents.
- Decreased average bug resolution time by 50% through systematic regression testing protocols and the creation of 500+ detailed test cases, proactively identifying over 200 edge cases pre-release using unit testing methodologies.

Lead Software Developer/Executive Board Member | Akomapa Health March 2025 - Present

- Architected and launched an end-to-end health management system utilizing a service-oriented architecture, scaling to serve 500+ global users across 15 countries with 99.7% uptime and significantly impacting non-communicable disease awareness.
- Boosted user engagement by 75% and increased monthly active users from 800 to 1,400 by leading a team of 3 developers to deliver 12+ feature releases, ensuring optimal platform performance through RESTful API development.

Machine Learning Engineer - Intern | Cita Marketplace.com Summer 2022 & 2023

- Deployed an ML recommendation engine that personalized product suggestions for over 50,000 monthly active users, contributing to over $250K in additional revenue and a 15% conversion rate increase.
- Improved search relevance accuracy from 68% to 88% and boosted customer retention by 18% by building predictive models analyzing 2M+ data points, reducing ML feature deployment time from 6 weeks to 4 weeks.
- Contributed to CI/CD practices for model deployment using NoSQL databases, ensuring faster iteration cycles and integration with existing systems.

Graduate Research Assistant | Oregon State University September 2021 - December 2023

- Engineered ML models using SVM, decision trees, and gradient descent optimization to achieve 92% prediction accuracy across 100,000+ data points, accelerating computational processing by 25%.
- Streamlined research workflow processing 50GB+ datasets by building automated Python pipelines that cut data preprocessing time by 40%, enabling the production of 1 published research paper.
- Leveraged deep learning techniques to analyze complex datasets, contributing to advanced PCA and t- SNE visualization and exploring Elasticsearch for efficient data retrieval.

PROJECTS

GradGPT.pro

- Leading development of an AI-powered platform to democratize college admissions guidance, reducing consultation costs by 95% and making premium guidance accessible to middle-income families across 4 countries.
- Building scalable microservices architecture using Node.js, Typescript, and React, implementing intelligent AI model routing with OpenAI GPT-4 integration for cost optimization and targeting 99.5% uptime.
- Generated 12K+ personalized essay reviews and research proposal feedback, helping 89% of users improve application quality through robust code development and deployment strategies.

DocuMind

- Architected intelligent document processing pipeline using PyTorch and FastAPI with custom OCR and NLP models, achieving 96.8% accuracy in structured data extraction.
- Automated workflows that previously required 200+ manual hours monthly by processing 10K+ unstructured documents including PDFs, images, and scanned forms.
- Implemented Infrastructure as Code principles for AWS deployments, ensuring scalability and reliability of the document processing system.

CodeExplain

- Built automated code documentation system using Azure OpenAI and AST parsing, generating comprehensive documentation for 5K+ lines of code across 15 repositories.
- Created architectural diagram generation feature producing Mermaid and PlantUML diagrams from code structure analysis.
- Integrated with CI/ CD pipelines to automatically generate and update documentation upon code commits, streamlining the development workflow.

GPU Memory Profiler for TensorFlow & PyTorch

- Contributed to GPU memory profiling tool optimizing deep learning workflows by visualizing model memory usage for TensorFlow and PyTorch.
- Enhanced debugging, training efficiency, and memory usage visualization by identifying bottlenecks, aiding ML practitioners in improving model performance.
- Utilized advanced Python libraries for data analysis and visualization to present performance metrics clearly.

SKILLS

- Programming Languages: Proficient in Python, Javascript, C++, SQL, Java, Typescript, Go, and Rust, with extensive experience in Node.js for backend development and RESTful API design.
- Cloud & DevOps: Skilled in CI/CD integration with Jenkins, Docker, Kubernetes, GitHub Actions, and Terraform for building robust and automated deployment pipelines across AWS and Google Cloud.
- Databases & Tools: Expertise in Git, Nextjs, Redis, Playwright, Tailwind CSS, Keras, Pandas, ScikitLearn, Matplotlib, PyTorch, Tensorflow, Selenium, Flask, Django, Express, PostgreSQL, MongoDB, React, SciPy, Streamlit, YOLO, FastAPI, OpenAI API, Azure OpenAI, Mermaid, and PlantUML.
- Advanced Concepts: Deep Learning, SVM, Decision Trees, Gradient Descent Optimization, PCA, t-SNE for data analysis and model engineering, coupled with strong unit testing and NoSQL database management.